03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 12

(JCD 8.3 - 8.4)

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Previous lecture

- For nuclear fusion to happen, quantum tunneling is required
- We defined the cross section of a reaction in terms of energy:

$$\sigma(E) = \frac{S(E)}{E} \exp\left(-\frac{b}{\sqrt{E}}\right),$$

- The Gamow energy E_G is defined as

$$\sqrt{E_G} = b = \frac{2\pi\, e^2 \sqrt{m_u}}{\sqrt{2}\hbar\, 4\pi\, \epsilon_0} Z_1\, Z_2\, \sqrt{A}$$

- The released energy (related to nuclei binding energy) is

$$Q = [\Delta m(A) + \Delta m(a) - \Delta m(Y) - \Delta m(y)]\, c^2$$

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Outline

The average reaction rate

Energy generation rate

Nuclear reactions

Hydrogen burning

Summary

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Simple case

- Assume the reaction $A(a, y)Y$
- Assume all particles A to be stationary and particles a to move at a constant velocity v
- Particles have number densities n_A and n_a
- Consider a box of side area dS and length dl
- Remember: cross section relates to the area around a particle within which a reaction can occur
- You can show (verify!) that the reaction rate per unit volume can be expressed as

$$r_{aA} = n_A \, n_a \, \sigma \, v$$

True average reaction rate

- In reality, velocity or cross section will not be constant, but rather a distribution - Maxwell-Boltzmann (see eg statistical physics). We also use v for the relative velocity between particles.
- We then express reaction rate as

$$r_{aA} = n_A \, n_a \, \langle \sigma \, v \rangle$$

- Small exception: if the reaction occurs between two particles that are the same, we "double counted" particles in our argument earlier
- In such a case, reaction rate becomes

$$r_{AA} = \frac{1}{2} n_A^2 \, \langle \sigma \, v \rangle$$

Velocity distribution

- The velocity distribution is expressed as

$$\langle \sigma\, v \rangle = \int_0^\infty \sigma(E)\, v\, f(v)\, dv = \int_0^\infty \sigma(E) \sqrt{\frac{2E}{m}}\, P(E)\, dE$$

- Remember that energy is expressed as the kinetic energy using the relative velocity and the reduced mass
- The Maxwell distributions are:

$$f(v)dv = 4\pi \left(\frac{m}{2\pi\, k\, T}\right)^{3/2} \exp\left(-\frac{m\, v^2}{2\, k\, T}\right) v^2\, dv$$

$$P(E)dE = \frac{2}{\sqrt{\pi}} \frac{E^{1/2}}{(k\, T)^{3/2}} \exp\left(-\frac{E}{k\, T}\right) dE$$

Gamow peak

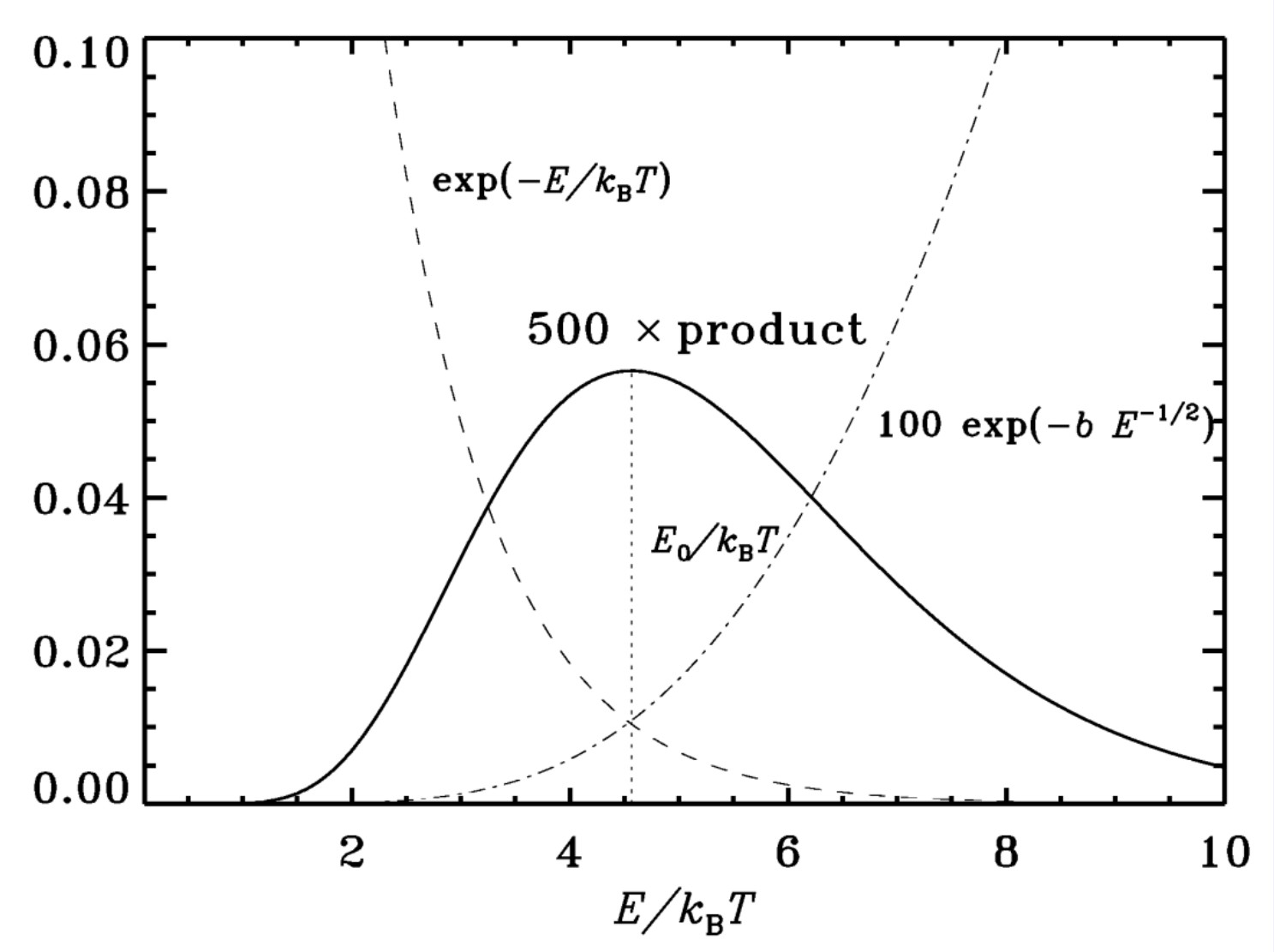
- Using all the definitions above, we can now show that the average reaction rate is proportional to

$$\langle \sigma\, v \rangle = \left(\frac{8}{m\pi}\right)^{1/2} \frac{1}{(k\,T)^{3/2}} \int_0^\infty S(E) \exp\left(-\frac{E}{k\,T} - \frac{b}{\sqrt{E}}\right) dE$$

- In the exponential, we now have one term decreasing with E and one term increasing with E
- There is a narrow range of energies around the maximum of this function. This range is called the Gamow peak.
- We can easily show that the maximum of the Gamow peak happens at an energy

$$E_0 = \left(\frac{b\,k\,T}{2}\right)^{2/3}$$

Gamow peak for reaction between 2 protons at $T = 15\,\mathrm{MK}$



(Figure credit: JCD)

Gamow peak increases strongly with temperature

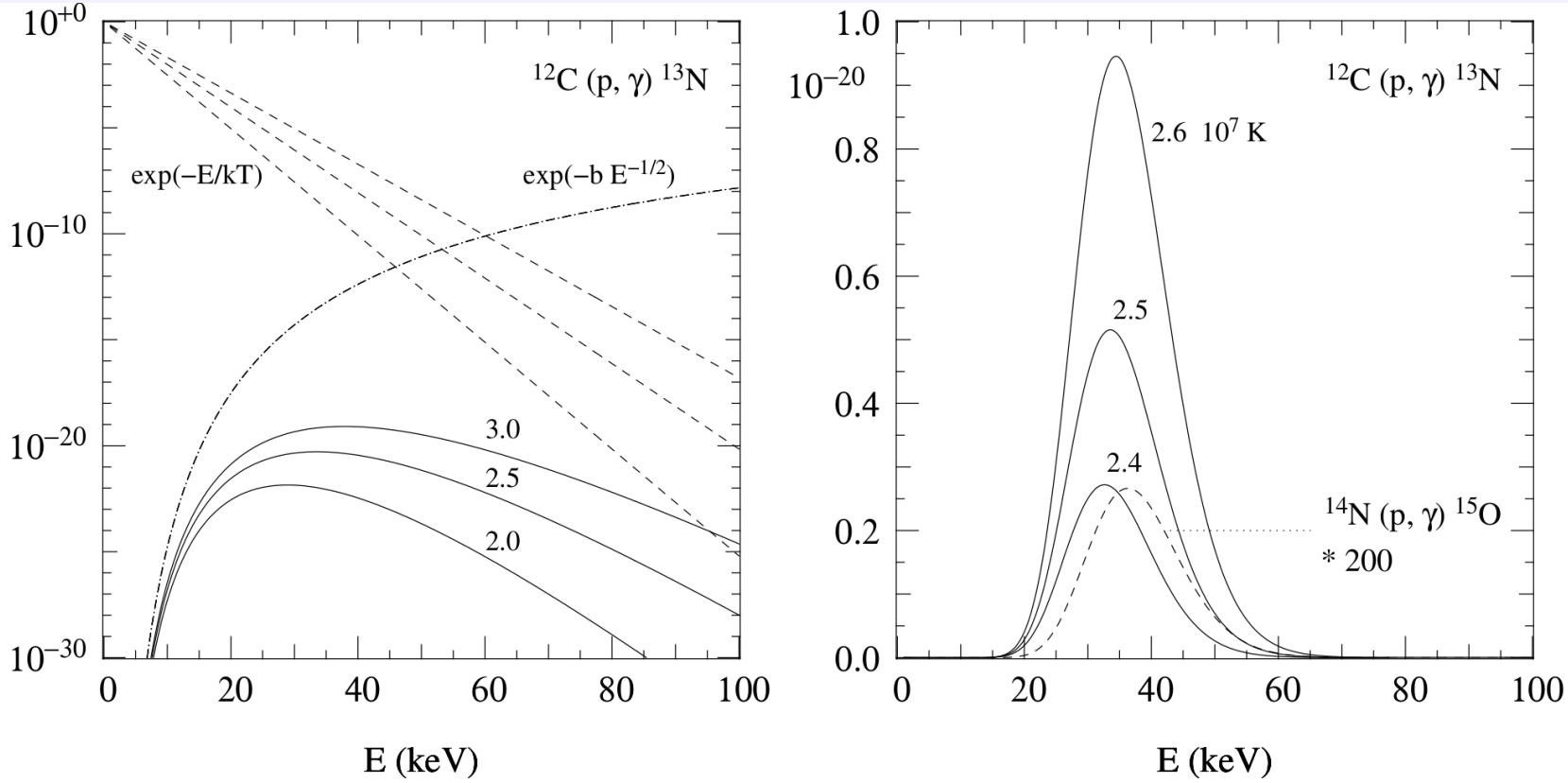


Figure 6.4. Example of the Gamow peak for the $^{12}C(p,\gamma)^{13}N$ reaction. The left panel shows as dash-dotted lines the tunnelling probability factor, $\exp(-b/E^{1/2})$, and as dashed lines the tail of the Maxwell distribution, $\exp(-E/kT)$, for three values of temperature: $T = 2.0\times10^7$ K (lower curve), 2.5×10^7 K (middle) and 3.0×10^7 K (upper). The solid lines show the product of these two factors, $f(E)$ as in eq. (6.21), labelled by $T_7 \equiv T/10^7$ K. Note the enormous range of the vertical log-scale. To appreciate the sharpness of the Gamow peak, and the enormous sensitivity to temperature, the right panel shows $f(E)$ on a linear scale for $T_7 = 2.4$, 2.5 and 2.6. The dashed line is the Gamow peak for the $^{14}N(p,\gamma)^{15}O$ reaction for $T_7 = 2.4$, multiplied by a factor 200.

(Figure credit: Onno Pols)

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Energy generation rate

- We now have all the ingredients for the average energy generation rate per unit mass, ϵ.
- It is related to the average reaction rate, the energy released and the density (note we used number densities in reaction rate)

$$\epsilon_{Aa} = \frac{r_{Aa}\, Q_{Aa}}{\rho}$$

- The total energy generation rate is then the sum of the energy generation rates from all involved reactions
- The change in mass fraction of the involved element can then be expressed as

$$\frac{dX_A}{dt} = -\langle \sigma\, v \rangle\, n_a X_A = -\frac{X_A}{\tau_{Aa}}$$

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Let's chat particles.

- ${}^{A}_{Z}E$ where E is the element, A is the atomic mass (protons plus neutrons), and Z is the atomic number (protons)
- Z defines the element; elements with same Z, but different A are isotopes
- proton, p, positive electric charge
- neutron, n, no charge, same mass as proton
- electron, e^-, negative electric charge, negligible mass
- anti-particle: same mass, opposite charge, opposite lepton number
- positron, e^+, positive electric charge, negligible mass, antiparticle of e^-
- neutrino, ν, no charge, negligible mass
- antineutrino, $\bar{\nu}$, no charge, negligible mass, antiparticle of ν
- photon, γ, massless and chargeless

Conservation laws in nuclear reactions

- Mass - energy, $E = m\,c^2$
- Total electric charge
- Number of leptons and antileptons
  (e.g. electrons, neutrinos, positrons, photons, ...)
- Number of baryons and antibaryons
  (e.g. protons, neutrons, ...)

Common reactions in stars

- Beta decay
  - β^- decay, spontaneous reaction, weak interaction

    $n \rightarrow p + e^- + \bar{\nu}$ i.e. ${}^A_Z E \rightarrow {}^A_{Z+1} F + e^- + \bar{\nu}$

  - β^+ decay, needs energy ($m_n > m_p$), weak interaction

    $p \rightarrow n + e^+ + \nu$ i.e. ${}^A_Z E \rightarrow {}^A_{Z-1} F + e^+ + \nu$

  - Electron capture ("inverse beta decay")

    $p + e^- \rightarrow n + \nu$ i.e. ${}^A_Z E + e^- \rightarrow {}^A_{Z-1} F + \nu$

- Photon processes
  - Photon gets released: (p, γ), (n, γ), (α, γ)
  - Photon is captured and liberates a particle: (γ, α), (γ, n)

Examples of photon processes

- (p, γ): proton reacts with nucleus and produces a more massive particle and a photon:

$$ {}^{A}_{Z}E + p \rightarrow {}^{A+1}_{Z+1}F + \gamma $$

$$ \text{e.g. } {}^{12}_{6}\text{C} + {}^{1}_{1}\text{H} \rightarrow {}^{13}_{7}\text{N} + \gamma $$

- (α, γ): alpha particle (i.e. ${}^{4}_{2}\text{He}$ helium nucleus) reacts with nucleus and produces a more massive particle and a photon:

$$ {}^{A}_{Z}E + {}^{4}_{2}\text{He} \rightarrow {}^{A+4}_{Z+2}F + \gamma $$

$$ \text{e.g. } {}^{3}_{2}\text{He} + {}^{4}_{2}\text{He} \rightarrow {}^{7}_{4}\text{Be} + \gamma $$

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Hydrogen fusion

- For most of a star's life it will fuse hydrogen into helium. The process is summarised by:

$$4\,{}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2\,e^{+} + 2\,\nu_e$$

- Using the energy release equation from lecture 11, we find that the fusion of hydrogen into helium releases

$$Q = 26.7\,\mathrm{MeV}$$

- However, the neutrinos will directly escape from the star, including their energy, and thus we need the effective energy release Q_{eff} where the value will depend on the actual reactions involved.
- The reaction above does not happen straight this way, but is rather done through an elaborate process of other reactions - see lecture 13: proton-proton chain, CNO cycle
- Each reaction process comes with its own effective energy release factor

Summary

- Average reaction rate: $r_{aA} = n_A \, n_a \, \langle \sigma \, v \rangle$
- Cross section-velocity distribution:

$$\langle \sigma \, v \rangle = \left(\frac{8}{m\pi} \right)^{1/2} \frac{1}{(k \, T)^{3/2}} \int_0^\infty S(E) \exp\left(-\frac{E}{k \, T} - \frac{b}{\sqrt{E}} \right) dE$$

- The Gamow peak is the narrow range of energies around the maximum of the exponential in this function.
- Energy generation rate: $\epsilon_{Aa} = \frac{r_{Aa} \, Q_{Aa}}{\rho}$
- Hydrogen fusion is summarised as $4 \, {}^1_1\mathrm{H} \rightarrow {}^4_2\mathrm{He} + 2 \, e^+ + 2 \, \nu_e$ where a total of 26.73 MeV is released

NEXT LECTURE: proton-proton chain, CNO cycle, later reactions, full set of stellar structure and evolution equations